Ball on Ball Action

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At a height of H above the ground, one billiard ball of mass m is stacked on top of another of mass $M > m$. Suppose that the balls are dropped simultaneously and with care, so that the line connecting the centers of the two balls remains vertical. After M makes contact with the ground and rebounds, momentum is transferred to m, causing it to bounce to a maximum height of H'. Our task it calculate H'. Assume that all collisions (ball-ground and ball-ball) are elastic and neglect air resistance, so that the total energy of the two-ball system is conserved. Furthermore, assume that the radii of the balls are negligibly small compared to H.

When M makes contact with the ground, the speed of M and m are $v = \sqrt{2gH}$, where g is the gravitational acceleration, and v is defined to be positive. Imagine a tiny gap between balls M and m. Without being in contact with m, ball M rebounds from the ground with a speed v. There is then a short time between the rebound of M and the collision with ball m. In the rest frame of the ground, the momenta of the balls immediately before the collision are $+Mv$ and $-mv$, where the positive and negative signs refer to velocities that directed away from and toward the ground, respectively. At this point, it is useful to work in the center-of-mass (CM) frame of the two balls, after M rebounds from the ground, but before it collides with ball m. The CM velocity is

$$V = \left(\frac{M-m}{M+m}\right) v \ . \tag{1}$$

The momenta of M and m in the CM frame before the collision are then $M(v-V)$ and $-m(v+V)$, the sum of which, of course, vanishes. Immediately after the collision, the velocities of M and m in the CM frame are denoted, respectively, by u' and w', so that the conserved total momentum is

$$Mu' + mw' = 0 \ . \tag{2}$$

Conservation of kinetic energy provides a second relation that allows us to uniquely determine u and w:

$$Mu'^2 + mw'^2 = M(v-V)^2 + m(v+V)^2 \ . \tag{3}$$

After solving eq. (2) for u, substituting the result in eq. (3), and carrying out some minor algebra, we find that

$$w' = 2\left(\frac{M}{M+m}\right) v \ , \tag{4}$$

and, on backsubstituting this result into eq. (2), we have

$$u' = -2\left(\frac{m}{M+m}\right) v \ . \tag{5}$$

In the rest frame of the ground, the post-collision velocities are

$$w = w' + V = \left(\frac{3M-m}{M+m}\right) v \tag{6}$$

and

$$u = u' + V = \left(\frac{M-3m}{M+m}\right) v \ . \tag{7}$$

The maximum height, H', attained by m following its recoil is given by

$$\frac{H'}{H} = \left(\frac{w}{v}\right)^2 = 9\left(\frac{M - m/3}{M + m}\right)^2 \ . \tag{8}$$

Note that the maximum possible value of $H'/H = 9$ occurs in the limit as $m \to 0$.